FarField: Grounded Jumps and Branch-Scoped Evolution

Proposed framework · Real observations inform future research and reusable capabilities

PAPERFUTURE / FRAMEWORK v0.1

Research intent + initial materials | Inspect → provisional questions, bottlenecks & unknowns

Stable intent and constraints; revisable scientific subquestions

A Mechanism-guided jumps

GROUND · TRANSFER · INVESTIGATE

Technical trajectory & bottleneck

Model drafts; sources check changes and limits

Prior methods → Mechanism shift → Bottleneck

A testable mechanism bridge

Source mechanism → shared structure → target

State assumptions, alternatives and failure conditions

Underexplored, conditionally valuable, testable

Landing inquiry & discriminating test

Check related methods, negative results and scope

Design a cheap test; keep implementation open

Investigate further / revise the landing / defer

Jump when needed; refine locally using real findings

Idea

B OpenWorld: learn through action

PREDICT · ACT · OBSERVE · RESPOND

Expectations & discriminating predictions

Record alternatives, interventions and predictions

Explore freely; freeze the protocol for confirmation

Build capabilities & execute

Write methods, access data, construct measurements

Independent native evaluator + code and raw results

Real environment and observations ≠ internal rollout

Validate measurements & respond

Check implementation, intervention and test validity

Cite results: retain / revise / test further / pause

Prediction error alone does not refute a hypothesis

Method performance | Separate tests | Mechanism evidence

Results

C Branch-scoped evolution

CONDITIONAL MEMORY & EXECUTABLE SKILLS

Conditional research memory

Findings, failure boundaries and unresolved questions

Source observations, scope and evidence dependencies

Executable skill versions

PLANNED

Reusable analysis code and experimental procedures

Local trial → validate → reuse / deactivate

Evolve research capabilities; keep shared rules fixed

Fork a fixed snapshot; evolve separately

Parent · learning bundle v_t

Branch A · memory / skills | Branch B · memory / skills

Shared read-only sources; isolated learning updates

Evidence-linked memory and skills guide the next question, experiment or jump

Retracted evidence triggers dependency review; retaining a justified belief is allowed

Shared foundation | Literature & data · Isolated execution · Independent evaluation · Event ledger & dependencies · Budget

Research objective Valuable directions, independently validated outcomes, and better subsequent research

Implementation boundary Solid: existing foundations requiring further work. Dashed: executable skill lifecycle and frozen learning snapshots to be completed.

Skills currently remain prompt advice; the full loop, prediction calibration and research benefits are not yet established.